1. Boolean F4/F5/F6 and native generalized roots

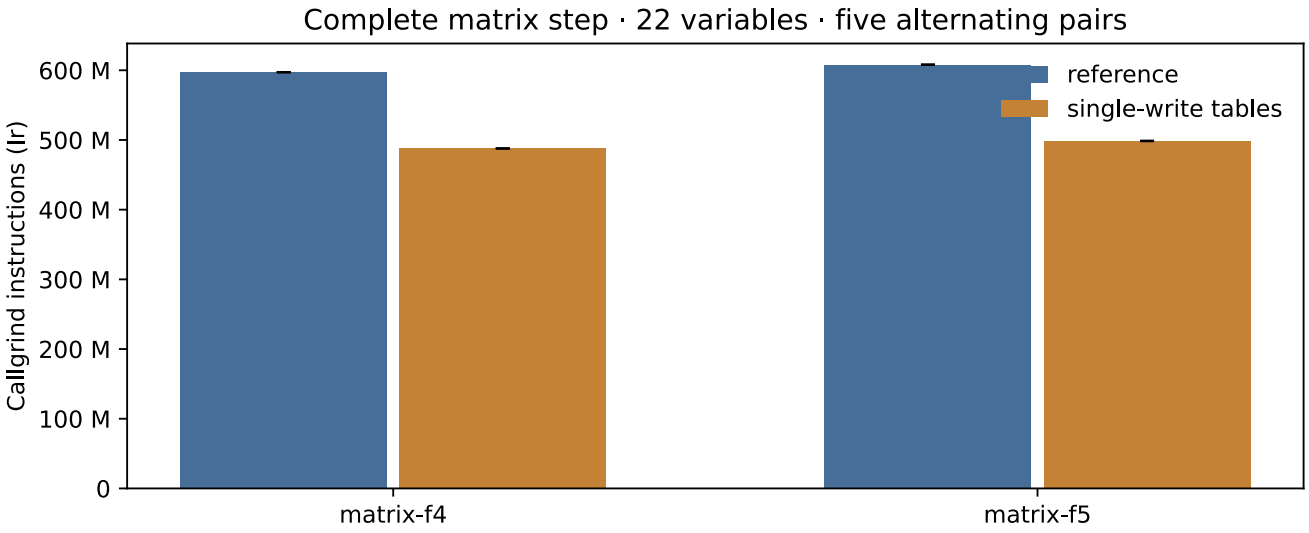
Instruction measurements, 2026-10-18. Native runtime qualification is separate.

The opt-in single-write Gray-table builder reduces complete matrix-F4/F5 instruction work on the linked 22-variable fixture, with identical reduced rows, rank and word-XOR accounting. These are Callgrind instruction results, not admitted native runtime speedups. The production default is unchanged.

The missing bounded generalized-rootfinding execution path is now native Rust: certified prime/binary/odd-extension fields; polynomial, affine-linearized and partial rational-composition domains; complete enumeration, component MITM and Boolean Gray dispatch; and a complete S3 separator retaining all original eight coordinates. Ten independent tests pass, including poles, repeated and inseparable roots, GF(2^83) extraction, a wide prime certificate, and literal eight-coordinate separator enumeration. CLI fixtures return exactly two Boolean roots and eight separator tuples. This completes these implementation contracts; it does not establish the old hard-family 8x research target.

2. Complete matrix step

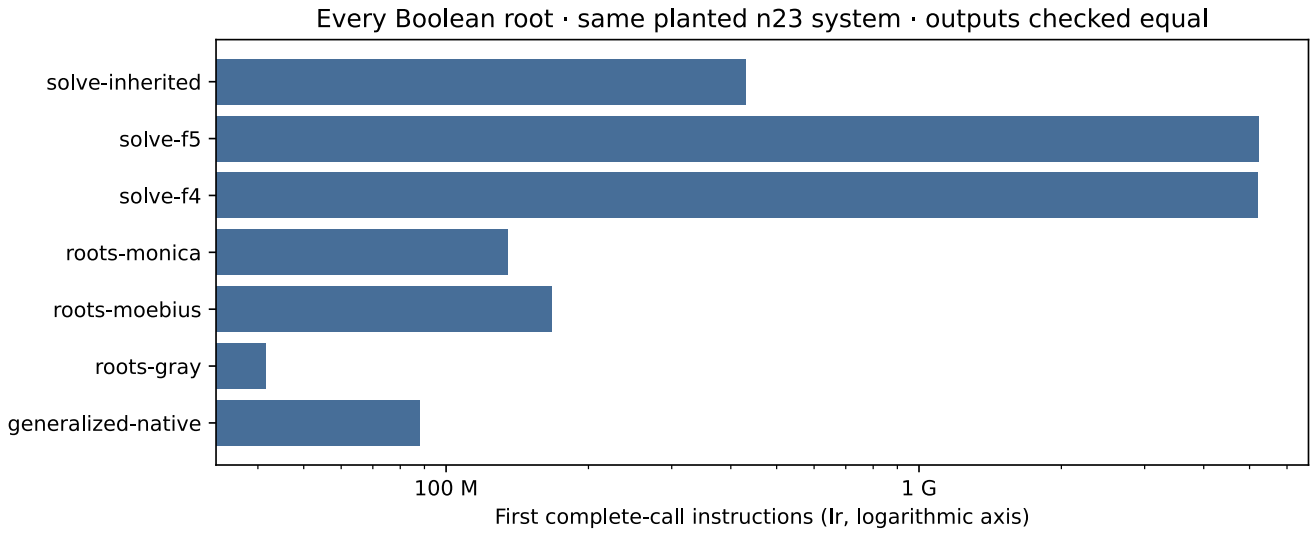
K1 n23, 22 Boolean variables, degree 4. Five alternating pairs; exact output and XOR accounting. Callgrind 3.19, merged workers, simulated AVX2. Fresh-seed and scaling records are retained in summary.json.



Engine	Reference Ir	Candidate Ir	Reduction
matrix-f4	597,267,029	487,814,530	18.326%
matrix-f5	607,964,549	498,549,578	17.997%

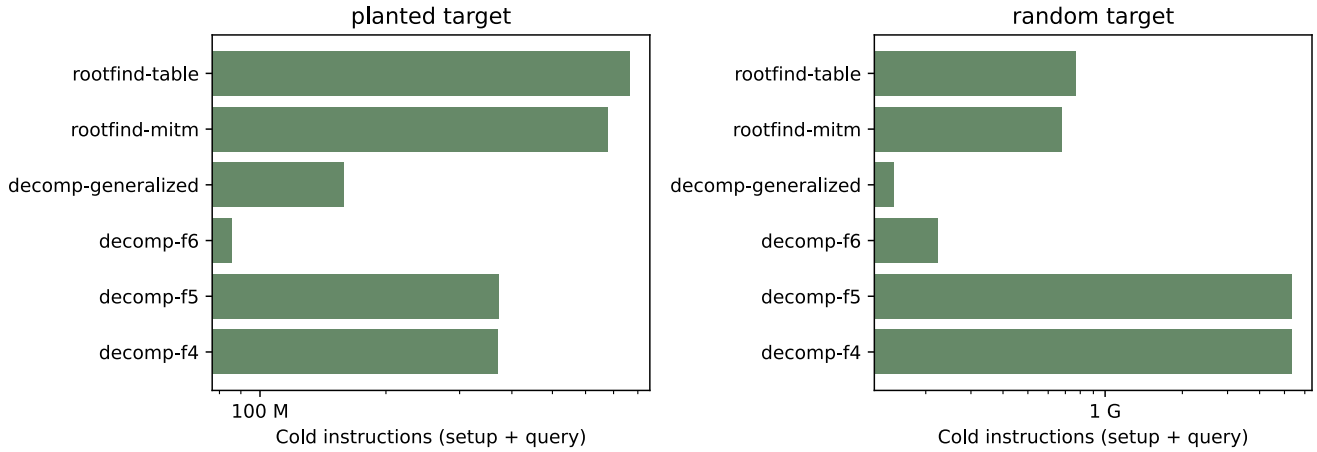
3. Every Boolean root

First complete call on the same n23 planted system. All original roots checked equal. The general API includes certification and normalization; direct Gray enumeration remains cheaper. Generic F4-basis profiling timed out on this fixture.



4. Cold same-target decomposition

One planted and one random n23 target. Table setup is added once; online query data remain separate. F6 and the generic solver can return different verified witnesses for the same verdict. No IC/rho result follows.



5. Reproduction and limits

Source: REPORT.md, PROTOCOL.md, summary.json, analyze.rs and runtime.rs in research/f4_f5_boolean_perf_20261018. Raw member hashes and every failed attempt are retained. Ten native solver tests and CLI fixtures pass; full-library status is in VALIDATION.md. Magma, msolve, FGb, Singular and PolyBoRi were not available. Native wall, external fastest, hard all-roots 8x and whole-pipeline claims are not established.